Production regions as decay-like triangles: plan

Goal. Evaluate the 579 iterated-integral generators, and hence the 84 finite remainders, in the three production regions $x > 1$, $y > 1$, $z > 1$ (the other two variables negative). We use the same differential equation and the same numerical method as in the decay region, and the same series pipeline. Crossings of the partons come later.

1. Bounding the regions. In region $A > 1$ the large variable is the energy. We write $A = 1/v$ and the other two as $-\hat{b}/v$, $-\hat{c}/v$, so that $\hat{b} + v + \hat{c} = 1$ with $\hat{b}, \hat{c}, v > 0$. The region becomes a triangle, and v is always the inverse energy. For example, for $z > 1$: $x = -w/v$, $y = -u/v$, $z = 1/v$, and $(X, Y, Z) = (w, u, v)$ plays the role of (x, y, z). These are the same variables as the continued-region charts of the boundary-constant transport.

2. The alphabet closes. Substituting the map into the twenty letters, every letter becomes a constant times a power of v times a letter of the same alphabet in (X, Y). The odd letters need the continued root $\sqrt{xyz} \to -\sqrt{XYZ}/Z^2$, and then

$$d\log W_7 \to d\log W_7' + d\log W_8', \qquad d\log W_8 \to -d\log W_8'.$$

For $z > 1$ the even letters map as follows:

W_1	W_2	W_3	W_4	W_5	W_6	W_{11}	W_{12}	W_{13}	W_{14}	W_{15}	W_{16}
W_1'/v	W_2'/v	$1/v$	W_5'/v	W_4'/v	W_6'/v	W_{12}'/v^2	W_{11}'/v^2	W_9'/v^2	W_{18}'/v^2	W_{17}'/v^2	W_{10}'/v^2

So in the region $dF = \sum_j d\log W_j' \, M_j' F$, with $M_j' = \sum_k c_{kj} M_k$ and c_{kj} integers. It is the same system on a triangle, and the same integrator applies.

3. The continuation sits in the boundary values. The corner $X, Y \to 0$ of the region's triangle is the attached decay corner, approached from the other side: c_1 for $z > 1$, c_2 for $x > 1$, c_3 for $y > 1$. Going from one side to the other is a half-turn around the corner. In the blow-up chart variables of that corner the Feynman prescription gives

$$\log t \to \log t - i\pi, \qquad \log B \to \log B + i\pi.$$

Chen's formula across the corner then reads

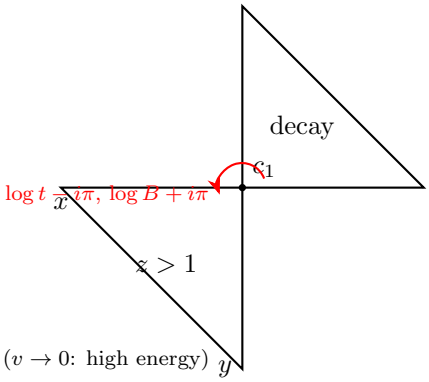
$$\mathrm{II}[a_1 \dots a_n](P) = \sum_k \mathrm{II}[a_1 \dots a_k](P^*) \; \mathrm{II}'_{P^* \to P}[a_{k+1} \dots a_n],$$

where:

- the values at the corner, $\mathrm{II}[\dots](P^*)$, are the logarithmic part of the decay expansion at the corner with the substitution above. They are polynomials in $i\pi$ (and MZVs at c_2, c_3), and in general nonzero;

- the remaining factors are iterated integrals in the new variables, starting at the corner, so they vanish there (regularised).

At the level of the whole system: the start values at the new corner are these polynomials instead of zero, and the connection is M'. Inside the triangle nothing is crossed, so no $i0$ is needed along the paths.



4. What we compute. For each region:

- the integer letter map c_{kj};
- the connection M', in the orthonormal gauge of the evaluator;
- the values at the corner from the ITER series at the attached decay corner (exact);
- the series start from the corner, carried to a hub $P_0' = (0.3, 0.3)$ of the region's triangle, and stored;
- evaluation of the generators and of the remainders. The rational functions are evaluated at the physical point, and $r = -\sqrt{|xyz|}$.

Checks.

1. The generators at points of the region: the region evaluator against a direct integration from the decay hub P_0, crossing $x = 0$ and $y = 0$ from above. These are two independent continuations.
2. The leading-colour layer against the production-channel files of the 2024 planar ancillary.
3. Later: the series at the corners of the region (threshold and high energy), whose constants are already known.